INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

## Degree: BCA, Stream: BCA

## Term – I , Fourth Semester Examination, February- 2025

## Subject Code: BCACC401

## Subject Name: Database Management Systems

Full Marks: 30 Set No: 1 Duration: 1 Hour

**Part - A**

**Answer any 5 questions out of 6**

**Each question carries 2 marks (2 × 5)**

1. What is a Candidate Key? **2**

2. What is an Alternate Key? **2**

3. What is the full form of SQL? **2**

4. Which type of language is SQL? **2**

5. What is DBMS? Give example. **2**

6. What is RDBMS? Give example. **2**

**Part - B**

**Answer any 2 questions out of 3**

**Each question carries 5 marks (5 × 2)**

7. All primary keys are super keys but the vice versa is not true. Justify the statement with an example. **5**

8. Justify the utility of foreign keys in establishing relationships among the tables with an example. **5**

9. Why SQL is considered as a non procedural and structured language? **5**

**Part - C**

**Answer any 1 question out of 2**

**Each question carries 10 marks (10 × 1)**

10. Create the following table using SQL. Student ( Roll- Number, Name- Varchar, DOB-Date, 10thMarks- Float Number,12thMarks- Float Number). Insert 2 records as per your own wish. **10**

11. Design an ERD for a hospital showing all different types of attributes , relationships and participation constraints. **10**

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Full Marks: 30 Set No: 2 Duration: 1 Hour

**Part - A**
**Answer any 5 questions out of 6**
**Each question carries 2 marks** (2 × 5)

1. Compare any two difference between file management system and DBMS. 2

2. Define the term data inconsistency. 2

3. How data integrity enforced in DBMS? 2

4. Compare between logical and physical level? 2

5. Illustrate derived attribute with a suitable example. 2

6. Retrieve ISBN and Title of all novel book, which are published by Laxmi publication from the bellow relation: 2

   BOOK (ISBN#, Title, Price, Category, Pulisher_Name, PID)

**Part - B**
**Answer any 2 questions out of 3**
**Each question carries 5 marks** (5 × 2)

7. Demonstrate various type of attribute in E-R Model. 5

8. EMPLOYEE (EID#, ENAME, SALARY, CITY, DNO) 5

   DEPARTMENT (DNO#, DNAME, DLOC)

   Retrieve all the employee name and there corresponding department those who work in Kolkata and got the salary more than Rs. 20000.

9. Explain various type of data integrity. 5

## Part - C

**Answer any 1 question out of 2**

**Each question carries 10 marks** (10 × 1)

10. Create the following table using SQL commands. Employee ( EmployeeID- Number, Name- Varchar, DOB-Date, Dept- Varchar(10), Designation Varchar(10),Salary- Float Number). Insert 2 records as per your own wish. **10**

11. Construct an ERD for an IT training group database that will meet the following condition: The Company has 12 instructors and can handle 100 trainees for each training session. The company offers 5 advanced technology courses, each of which is taught by a team of 2 or more instructors. Each instructors is assigned to a maximum of two teaching teams or may be assigned to do research. Each trainee undertake one advanced technology course per training session. **10**

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## Subject Code: BCACC401

## Subject Name: Database Management Systems

Full Marks: 30 Set No: 3 Duration: 1 Hour

**Part - A**
**Answer any 5 questions out of 6**
**Each question carries 2 marks** (2 × 5)

1. Mention two (2) advantages of DBMS approach over conventional file based system. 2

2. Define the term data redundancy 2

3. What do you mean by integrity constraints 2

4. Define the term database schema and database instance with an example. 2

5. Differentiate between simple attributes and composite attributes with example. 2

6. List the conditions of defining primary key. 2

**Part - B**
**Answer any 2 questions out of 3**
**Each question carries 5 marks** (5 × 2)

7. Demonstrate various type of attribute in E-R Model. 5

8. Why SQL is considered as a non procedural and structured language? 5

9. Write a short note on 3 tier architecture of modern Database Management systems 5

**Part - C**
**Answer any 1 question out of 2**
**Each question carries 10 marks** (10 × 1)

10. Create the following table using SQL. Student ( Roll- Number, Name- Varchar, DOB-Date, 10thMarks- Float Number,12thMarks- Float Number). Insert 2 records as per your own wish. 10

11. Construct an ERD for an IT training group database that will meet the following condition: The Company has 12 instructors and can handle 100 trainees for each training session. The company offers 5 advanced technology courses, each of 10

which is taught by a team of 2 or more instructors. Each instructors is assigned to a maximum of two teaching teams or may be assigned to do research. Each trainee undertake one advanced technology course per training session.

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Subject Code: BCACC401

Subject Name: Database Management Systems

Full Marks: 30 Set No: 3 Duration: 1 Hour

**Part - A**
**Answer any 5 questions out of 6**
**Each question carries 2 marks** (2 × 5)

1. Mention two (2) advantages of DBMS approach over conventional file based system. **2**

2. Define the term data redundancy **2**

3. What do you mean by integrity constraints **2**

4. Define the term database schema and database instance with an example. **2**

5. Differentiate between simple attributes and composite attributes with example. **2**

6. List the conditions of defining primary key. **2**

**Part - B**
**Answer any 2 questions out of 3**
**Each question carries 5 marks** (5 × 2)

7. Demonstrate various type of attribute in E-R Model. **5**

8. Why SQL is considered as a non procedural and structured language? **5**

9. Write a short note on 3 tier architecture of modern Database Management systems **5**

**Part - C**
**Answer any 1 question out of 2**
**Each question carries 10 marks** (10 × 1)

10. Create the following table using SQL. Student ( Roll- Number, Name- Varchar, DOB-Date, 10thMarks- Float Number,12thMarks- Float Number). Insert 2 records as per your own wish. **10**

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